Structured Conversation Segment: Defining Conversational Minimal Logic Unit (CMLU) for LLM-based Chat Analysis at Scale

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Abstract

Revise at last (150 words)
As LLM-powered chat bots are developing dramatically, a growing volume of user-bot semi-structured data are being generated. This type of conversations are usually messy, disordered and with fast topic shifting which gives a big challenge for traditional NLP that trained on well-structured data (clean paragraphs, single-topic document). We proposed the **Conversational Minimal Logic Unit (CMLU)** as a topically-atomic, smallest unsegmented semantic unit - analogous to how sentences are the smallest unit over which grammatical relations close, CMLU are the minimal unit of conversational semantics, and **UFL**, a three-tier architecture (deterministic pre-segmentation, template matching, dual-prompt LLM analyzer) that extracts CMLUs at production scale. (add results summary)

1 Introduction

(1 pages)
A typical conversation between LLM-powered chat bot and user generates well formatted, repeating templates, lack of user's responses, and topic shifts. However, the most important character of this kind of conversation is **open-ended, session-less conversation steam** which distinguishes it from current session bounded dialogue datasets.

1. **Noise**: With the development of LLM chatbot: massive amount of repeating contents (templates), they are very noisy and harmful for doing topic classification and sentiment analysis.

2. **Complexity**: User chats contains a large portion of randomness, uncertainty, especially chatting with LLM powered chatbot, they usually have such features:

   - **Asynchronous**: conversations may not flow sequentially
   - **Lack of Participation**: the user usually prefers to say less when chatting with the chat bot which makes individual message have very little signal
   - **Multi-topic Drifting**: user's topic with chat bot can change faster without any logistics in between

3. **Current methods' shortage / Why existing segmentation fails here**: with inspection of all the current work on conversation segmentation and analysis, none of them dealing with the infinite content streaming problem, their shortage includes:

   - **Lack of Rubric**: do not have a ground truth rule for LLM-based pipeline to perform the segmentation
   - **Utterance Level Processing**: dialog segmentation and classification operate at the user utterance level
   - **Sequential Chat Only**: Segmentation assumed the sequential flow, lack of ability dealing with asynchronous dialog, and

Contribution:

1. We formalized the **Conversational Minimal Logic Unit (CMLU)** — topically-atomic, smallest unit of semantic for LLM-based analyzer to segment the conversation. CMLU generalizes boundary-based conversation segmentation to the open-ended, session-less dialogue setting

2. We proposed **UFL**, a three-tier architecture (deterministic pre-segmentation → template matching → dual-prompt LLM analyzer) that extracts CMLUs together with topic, subtopic, summary, sentiment, and confidence in a single LLM call per user-window

3. We proposed a **dual-path routing** (user vs. bot utterance streams) which identify the failure mode invisible in academic datasets: boilerplate bot content dominating topic signals.

4. We proposed a novel evaluation system that contains the following:

   - **Hungarian matching**: evaluate segments matching accuracy without constrained by position-aligned (dealing with asynchronous dialogue)
   - **F1 Metric redesign**:

5. **(TBD)** We validate at production scale (130K users, 3M messages), achieving F1 = 0.952 on expert-labeled CMLUs — a 3× improvement over direct LLM prompting.

2 Related Work

(1 page)
(Method: , UPS)

1. Dialogue and topic segmentation:

   - **Bechmarks**: **TIAGE**: A Benchmark for Topic-Shift Aware Dialog Modeling." (The standard topic-shift benchmark for open-domain dialog.); **Super-Dialseg**: A Large-Scale Dataset for Supervised Dialogue Segmentation. (9K dialogues, the largest labeled dialogue segmentation corpus.), **Doc2Dial**: A Goal-Oriented Document-Grounded Dialogue Dataset (Document-grounded, so adjacent to your setting but useful for positioning)
   - **DL segmentation methods**: Text Segmentation as a Supervised Learning Task. (Introduced Wiki-727K and the first strong neural supervised segmenter.); SECTOR: A Neural Model for Coherent Topic Segmentation and Classification (Joint segmentation + classification); Text Segmentation by Cross Segment Attention. (Strong transformer baseline for segmentation)

2. LLM-based conversation analysis (justify "LLM-in-the-loop is the right design choice." Pick papers that demonstrate):

   - **LLMs as annotators / classifiers**: ChatGPT Outperforms Crowd Workers for Text-Annotation Tasks. (Establishes LLM annotation as a credible methodology); ChatGPT: Beginning of an End of Manual Linguistic Data Annotation?
   - **Dialogue summarization**: News Summarization and Evaluation in the Era of GPT-3; SAMSum Corpus: A Human-annotated Dialogue Dataset for Abstractive Summarization.

3. Evaluation method of segmentation

3 Problem Definition

Formal (mathematical) definition of the problem: Unlimited Context Stream & CMLU (0.5 page)
Notation: Let $M = (m_1, m_2, \ldots)$ be a conversation stream — an ordered, potentially unbounded sequence of messages, each tagged with sender $s_i \in \{user, bot\}$ and timestamp t_i.

- **Open-ended Context Stream** defined as the non-session and non-constrained chat conversations Mhen sh between user and chatbot.

- **CMLU**: CMLU is a **Conversational Minimal Logic Unit** $\mathcal{C} \subseteq M$ is a subset of messages satisfying:

  1. **Topical atomicity:** there exists a topic t such that every $m \in \mathcal{C}$ is about t.
  2. **Maximality:** no $m \in M \setminus \mathcal{C}$ about topic t is excluded from $\mathcal{C}$.
  3. **Minimality under atomicity:** $\mathcal{C}$ cannot be partitioned into $\mathcal{C} = \mathcal{C}_1 \sqcup \mathcal{C}_2$ such that both $\mathcal{C}_1$ and $\mathcal{C}_2$ independently satisfy (i) and (ii) with distinct topics.

  Note that (i)–(iii) allow $\mathcal{C}$ to contain non-contiguous messages (indices in $\mathcal{C}$ need not form a contiguous range) and allow $m \in \mathcal{C}_1 \cap \mathcal{C}_2$ for distinct CMLUs $\mathcal{C}_1, \mathcal{C}_2$ (multi-membership).

4 Method

(2 pages)
Figure: UFL pipeline architecture diagram

5 Experiment Setup

(0.5 - 1 page)

6 Results

(1-1.5 page)
Figure: CMLU example (before/after)

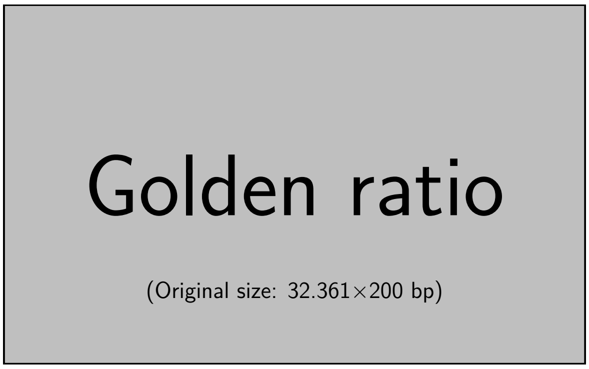


Figure 1: A figure with a caption that runs for more than one line. Example image is usually available through the `mwe` package without even mentioning it in the preamble.

7 Analysis

(1 page)

8 Conclusion

(0.25-0.5 page) Future work: Memory system: define a new "session management" method

9 Tips

9.1 Footnotes

Footnotes are inserted with the `\footnote` command.[1]

9.2 Tables and figures

See Table **??** for an example of a table and its caption. **Do not override the default caption sizes.**

As much as possible, fonts in figures should conform to the document fonts. See Figure 1 for an example of a figure and its caption.

Using the `graphicx` package graphics files can be included within figure environment at an appropriate point within the text. The `graphicx` package supports various optional arguments to control the appearance of the figure. You must include it explicitly in the LaTeX preamble (after the `\documentclass` declaration and before `\begin{document}`) using `\usepackage{graphicx}`.

9.3 Hyperlinks

Users of older versions of LaTeX may encounter the following error during compilation:

> `\pdfendlink` ended up in different nesting level than `\pdfstartlink`.

This happens when pdfLaTeX is used and a citation splits across a page boundary. The best way to fix this is to upgrade LaTeX to 2018-12-01 or later.

9.4 Citations

Table 1 shows the syntax supported by the style files. We encourage you to use the natbib styles. You can use the command `\citet` (cite in text) to get "author (year)" citations, like this citation to a paper by Gusfield (1997). You can use the command `\citep` (cite in parentheses) to get "(author, year)" citations (Gusfield, 1997). You can use the command `\citealp` (alternative cite without parentheses) to get "author, year" citations, which is useful for using citations within parentheses (e.g. Gusfield, 1997).

A possessive citation can be made with the command `\citeposs`. This is not a standard natbib command, so it is generally not compatible with other style files.

9.5 References

The LaTeX and BibTeX style files provided roughly follow the American Psychological Association format. If your own bib file is named `custom.bib`, then placing the following before any appendices in your LaTeX file will generate the references section for you:

```
\bibliography{custom}
```

You can obtain the complete ACL Anthology as a BibTeX file from `https://aclweb.org/anthology/anthology.bib.gz`. To include both the Anthology and your own .bib file, use the following instead of the above.

```
\bibliography{anthology,custom}
```

Please see Section **??** for information on preparing BibTeX files.

9.6 Equations

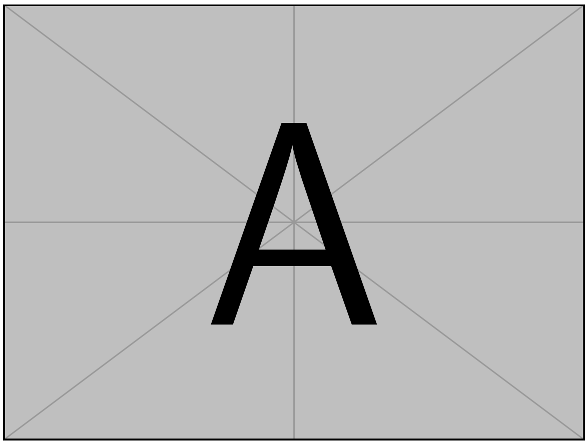
An example equation is shown below:

$$A = \pi r^2 \tag{1}$$

Labels for equation numbers, sections, subsections, figures and tables are all defined with the `\label{label}` command and cross references to them are made with the `\ref{label}` command.

This an example cross-reference to Equation 1.

[1] This is a footnote.



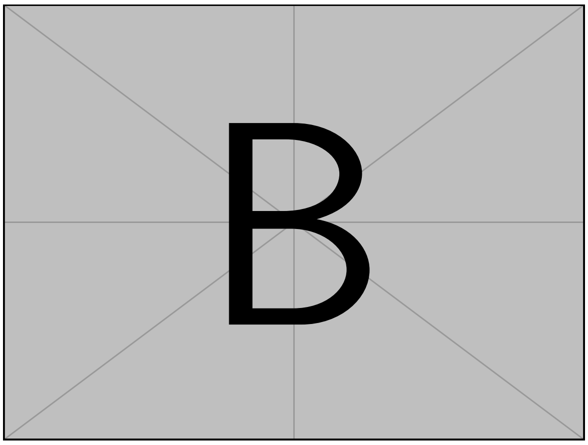


Figure 2: A minimal working example to demonstrate how to place two images side-by-side.

Output	**natbib command**	**ACL only command**
(Gusfield, 1997)	`\citep`	
Gusfield, 1997	`\citealp`	
Gusfield (1997)	`\citet`	
(1997)	`\citeyearpar`	
Gusfield's (1997)		`\citeposs`

Table 1: Citation commands supported by the style file. The style is based on the natbib package and supports all natbib citation commands. It also supports commands defined in previous ACL style files for compatibility.

9.7 Appendices

Use `\appendix` before any appendix section to switch the section numbering over to letters. See Appendix A for an example.

References

Rie Kubota Ando and Tong Zhang. 2005. A framework for learning predictive structures from multiple tasks and unlabeled data. *Journal of Machine Learning Research*, 6:1817–1853.

Galen Andrew and Jianfeng Gao. 2007. Scalable training of L1-regularized log-linear models. In *Proceedings of the 24th International Conference on Machine Learning*, pages 33–40.

Dan Gusfield. 1997. *Algorithms on Strings, Trees and Sequences*. Cambridge University Press, Cambridge, UK.

Mohammad Sadegh Rasooli and Joel R. Tetreault. 2015. Yara parser: A fast and accurate dependency parser. *Computing Research Repository*, arXiv:1503.06733. Version 2.

A Example Appendix

This is an appendix.